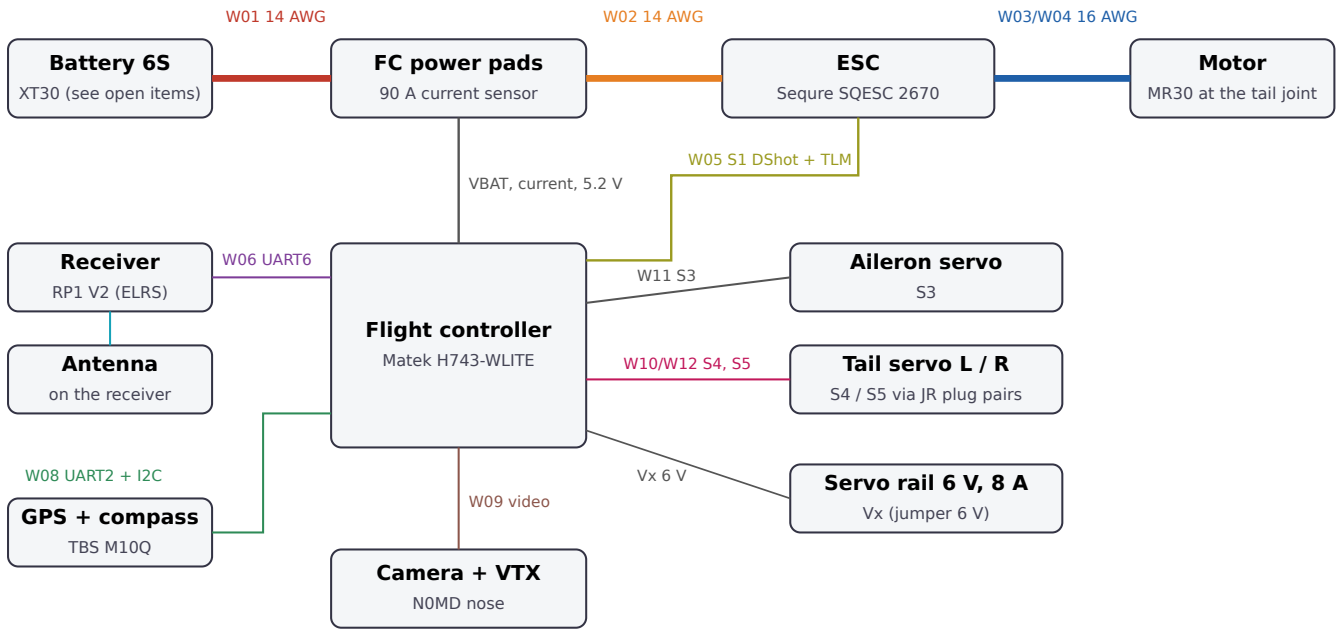
SkyG Speed 120 · Electrical integration manual

Power path, wire sizes and cut lengths, pinouts, routing, board settings, build order and tests. Cable lengths are measured on the 3D harness model of this build (the same routes the collision check uses). Generated from the design files; variant C7, flight nose N0MD.

> **Safety.** LiPo packs deliver over 100 A into a short. Work with the pack unplugged; first power-up through a smoke stopper or a current-limited supply; prop off for every bench test except the final static run; never charge inside the aircraft.

1. System overview



The battery feeds the Matek H743-WLITE's power pads first, so its 90 A current sensor measures everything; the ESC on top of the FC is fed from the same pads. The board makes 5 V for the electronics, 6 V for the servos and 9 V for the video. The motor plugs in at the tail joint (MR30) so the tail module comes off. The receiver and its antenna sit in the wing bay ahead of the FC.

2. Power budget and wire sizing

Cable	Circuit	Current A (pass / static)	Wire	Loop mm	R mOhm	Drop V	Loss W	dT K / pass	Connector rating
W01+W02	Battery to ESC (DC)	51 / 60	14 AWG	174	1.4	0.07 (0.3 %)	3.7	24	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	42 / 49	16 AWG	256	3.4	0.14 (0.6 %)	5.8	40	MR30: 30 A cont / 40 A for 1 min per contact

Currents from the propulsion model: 50.9 A at top speed (passes of <= 8 s), 60.0 A at full throttle on the ground. Phase RMS = sqrt(2/3) x DC. Temperature rise: copper only, no cooling, for one 8 s pass (an upper bound). Drop in % of the nominal 22.2 V.

Electronics: FC 5 V BEC 2 A continuous (FC, GPS, receiver about 1.2 A together); servo rail (Vx) 6 V / 8 A continuous for three micro servos; VTX rail 9 V / 2 A. Signal and servo wires: 26-28 AWG. Main DC path: 14 AWG (16 AWG would rise about 61 K per pass); motor phases: 16 AWG.

3. Cable schedule (cut list)

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W01	Battery lead	XT30 female (the pack plugs in), behind the pack under the GPS	Matek H743-WLITE power pads + / - (shared with W02)	red +, black -	14	57	**85**	XT30 solder cups + 3/16 in heat shrink; pads tinned, 3 mm stripped	Right lane (y +11), down in front of the FC to its power pads.
W02	ESC feed	Matek H743-WLITE power pads (after its 90 A sensor)	ESC BAT+ / BAT- (ESC on top of the FC)	red +, black -	14	30	**50**	Solder both ends; the ESC's capacitor across its BAT pads	Up the FC's front face in the right lane, over its top into the ESC's front.
W03	Motor phases, ESC side	Sequre SQESC 2670 motor pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the tail joint	3 x black (mark A/B/C)	16	27	**50**	Solder to the ESC's motor pads; MR30 solder cups + 3 mm heat shrink	From the ESC's rear, low on the left of the wing bay to the MR30.
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat shrink)	3 x black	16	101	**145**	MR30 solder cups; butt splices with 3 mm heat shrink	Through the joint's upper quadrant, along the F3 top, down to the motor bulkhead. Unplug the MR30 before pulling the tail module.
W05	ESC signal	ESC signal wires (signal, GND, telemetry)	FC S1 (signal, GND); telemetry to RX4	yellow signal, black GND, white telemetry	28	12	**40**	Solder at the ESC (or its supplied lead); 1-pin Dupont housings at S1	From the ESC's front down onto the FC's top, left lane. Never connect an ESC BEC (+) wire to the servo rail.
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on the bay floor ahead of the FC	FC UART6 JST-GH (5V, GND, RX6, TX6)	red 5V, black GND, yellow RP1-TX -> RX6, white RP1-RX -> TX6	28	5	**35**	Solder at the receiver; JST-GH 4-pin plug at the FC	Short: from the receiver's top to the FC's front face.
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC UART2 + I2C1 JST-GH (5V, GND, RX2, TX2, DA1, CL1)	6 x 28 AWG (the model shows the bundle as 4)	28	58	**95**	Supplied GPS plug; solder or GH plug at the FC	Under the GPS carrier; leave a slack loop so the hatch lifts off.

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W09	Video trunk (N0MD nose)	Nose joint plug (camera on the nose; VTX on the nose or, dome nose, under the hatch)	FC video pads (power, GND, camera in, VTX out)	red +, black GND, yellow camera video, white VTX video	28	204	**265**	4-pin JST plug pair at the nose joint; solder at the FC	Under the battery floor plate, up through the floor hole ahead of the F1/F2 bulkhead.
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	brown GND, red +6 V, orange signal	26	63	stock	Stock servo plug	Along the floor and through the tail joint; the plug pair separates the tail module. Needs a stock lead of at least the route length + 30 mm.
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	brown, red, orange	26	17	stock	100 mm stock extension (BOM line 35)	Along the floor to the FC's rear edge.
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	brown, red, orange	26	60	stock	Stock plug	Under the floor beside the servo, up behind the FC.

Cut length = route length on the model x 1.10 (slack, 1.15 for plugged leads) + the termination allowances, rounded up to 5 mm. Total cut length of made cables: 765 mm. Stock servo leads are used as supplied (W10 servo side, W11).

4. Connections and pinouts

Matek H743-WLITE

Pad / header	Connects	Setting / note
Power pads + / - (one pair)	W01 battery lead and W02 ESC feed on the same pads	14 AWG; the 90 A sensor sits between them and the board
S1: signal, GND	W05 ESC signal + GND (no + wire)	DShot300 bidirectional (SERVO1_FUNCTION 70)
RX4 pad	W05 ESC telemetry (optional)	AM32 serial telemetry, fallback rpm source (SERIAL6_PROTOCOL 16)
S3	W11 aileron servo	SERVO3_FUNCTION 4
S4 / S5	W10 + W12 tail servos L / R	SERVO4/5_FUNCTION 79 / 80 (V-tail)
UART6 (JST-GH): 5V, GND, TX6, RX6	W06 RP1 V2	CRSF: SERIAL7_PROTOCOL 23 with BRD_ALT_CONFIG 1
UART2 (JST-GH) + I2C1 (DA1, CL1)	W08 TBS M10Q	SERIAL3_PROTOCOL 5; compass on I2C1
Video pads (VTX power, GND, camera in, VTX out)	W09 camera + VTX	VTX rail 9 V by default (12 V jumper): check the VTX and camera accept it
UART8 (pin header)	DJI O4 Air Unit (O4 noses only)	SERIAL5_PROTOCOL 42, MSP DisplayPort

Sequre SQESC 2670: signal wires

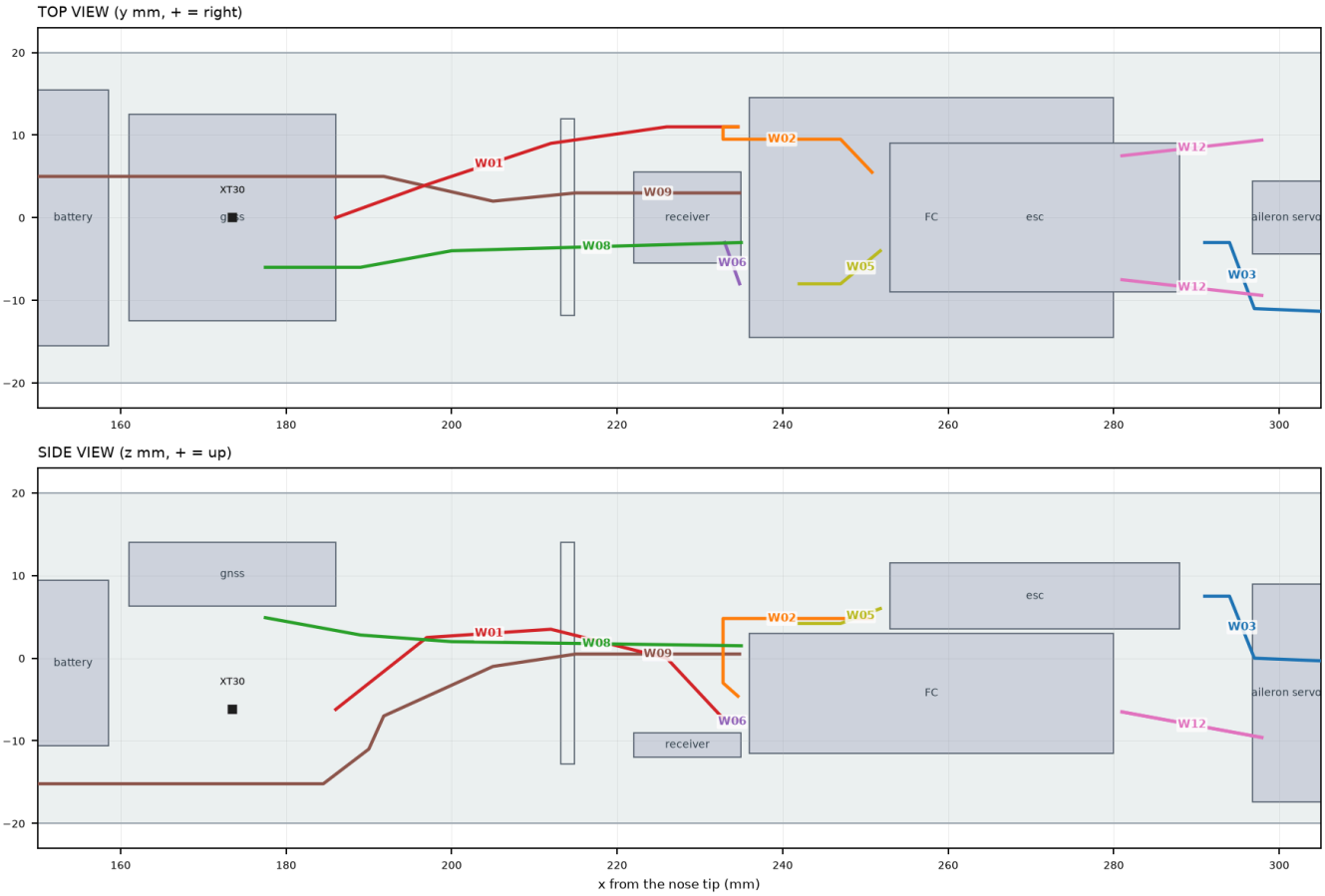
Pin	Wire	Use
Signal	yellow	W05 -> FC S1 signal
GND	black	W05 -> FC S1 GND
Telemetry (if fitted)	white	W05 -> FC RX4 (optional)
BEC + (if the ESC has one)	-	Not connected: the FC's own BECs power everything

> Never connect an ESC BEC (+) wire to the FC's servo rail or a 5 V pad.

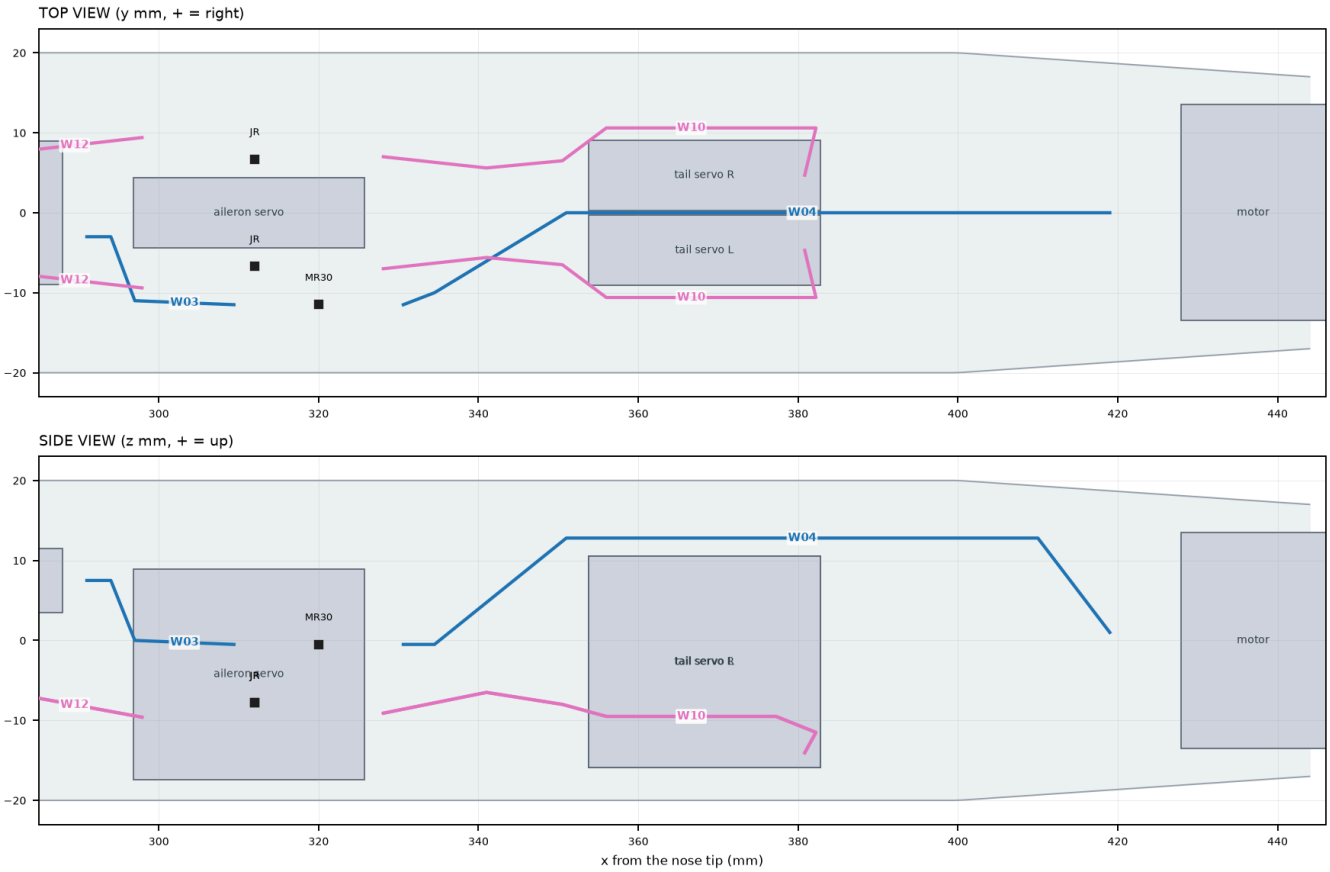
Connectors

Connector	Gender / pins	Note
XT30 (battery)	Female on the aircraft lead, male on the pack	Red +, black -; check polarity on the housing's chamfer
MR30 (motor)	Female on the ESC side, male on the motor side	Phase order free: set the direction in AM32, not by swapping wires
JR servo plug	Brown GND, red +6 V, orange signal	Same orientation on every header: signal towards the board edge marked S

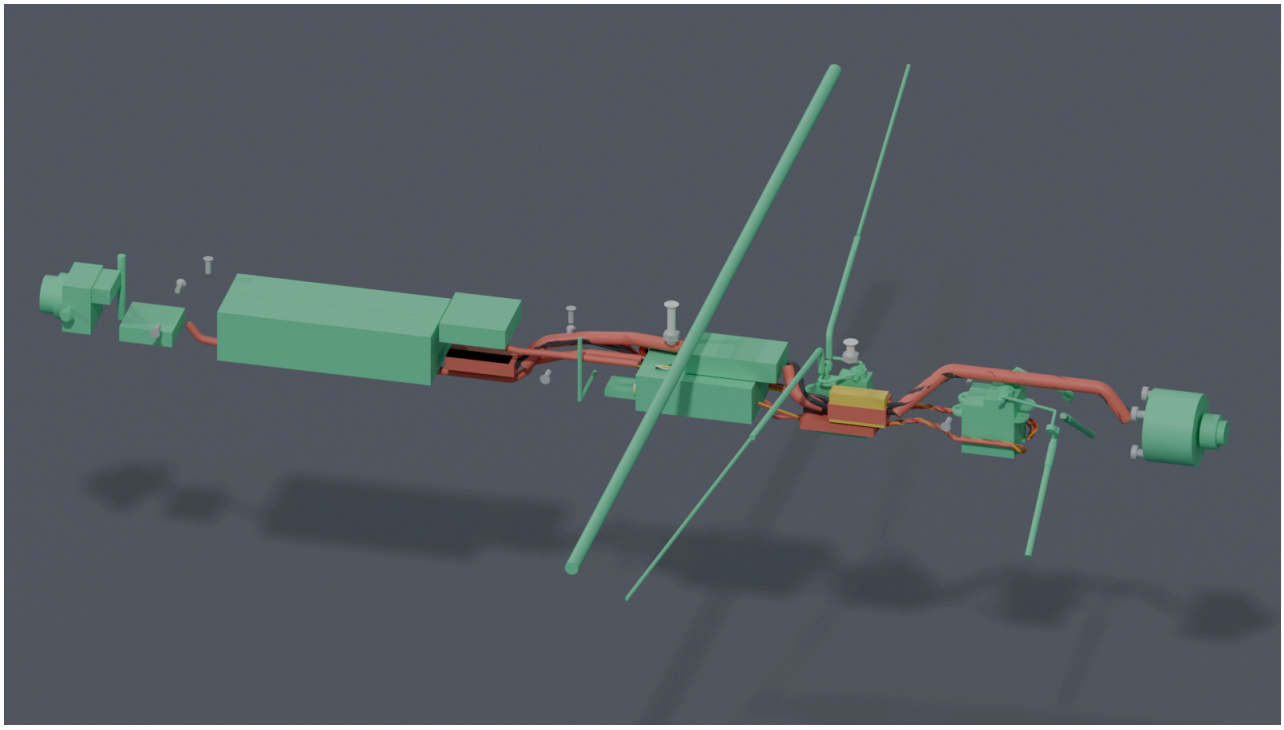
5. Routing



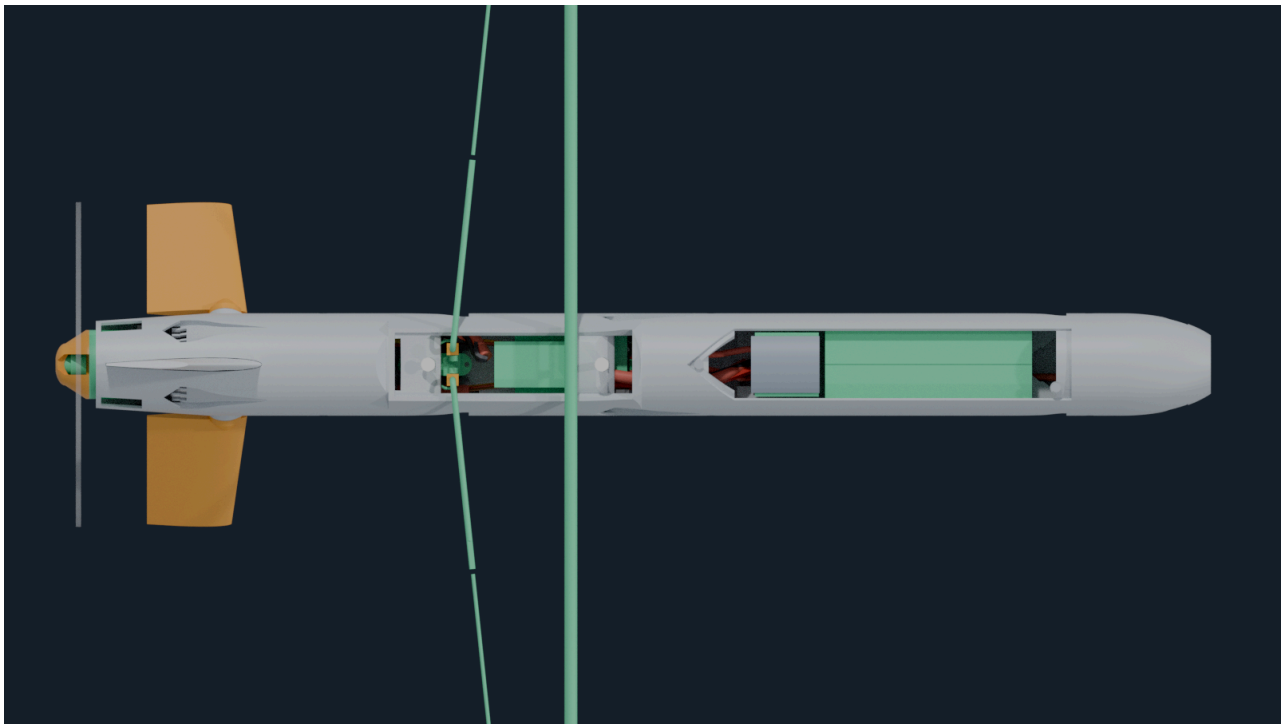
Front: battery bay, wing bay (ESC, FC, receiver).



Rear: aileron servo, tail joint, tail module, fin. Cable routes from the 3D harness model (x from the nose tip); squares are connector pairs. Every route is collision-checked against every printed part, component and cable.



Render: harness with the fuselage hidden.



Render: top view, fuselage cut open.

Lanes in the wing bay (y in mm): battery lead W01 and ESC feed W02 on the right (+11, over the FC's top at +9.5), video W09 at +3, GPS W08 at -3, ESC signal W05 on the FC's left top (-8), motor phases W03 low on the left. Receiver W06: short, straight to the FC's front.

6. Board and ESC settings

Item	Setting	Why
FC: Vx (servo rail) jumper	6 V	Emax ES9052MD: 1.3 kgf cm at 6 V. Rail 8 A continuous for the three servos.
FC: VTX rail jumper	9 V (default)	12 V only if the VTX needs it; check the camera's input range.
ESC (AM32): protocol	DShot300, bidirectional on	SERVO_BLH_MASK 1, SERVO_BLH_BDMASK 1, SERVO_BLH_OTYPE 5.
ESC (AM32): motor direction	Set on the bench	Pusher prop: the motor turns the prop so that it blows aft. Reverse in AM32 if wrong.
ESC (AM32): brake on stop	On	Holds the prop still for the belly landing.
ESC (AM32): motor poles	14	12N14P motor: SERVO_BLH_POLES 14 for correct rpm (the governor needs it).
FC: battery monitor	BATT_MONITOR 4, BATT_VOLT_MULT 21.0, BATT_AMP_PERVLT 66.7	The board's own divider and 90 A sensor: calibrate both against a meter (see the tests).
FC: rpm governor (Lua)	skyg_governor.lua, GOV_ parameters	Holds the prop rpm and the battery current limit; prove it on the thrust stand first.

7. Build order

1. Bench, before anything goes into the fuselage: flash ArduPilot (MatekH743-bdshot), load the parameter file and the governor script, set the Vx jumper to 6 V.
2. FC: solder W01 and W02 together on the power pads (14 AWG: 3 mm strip, pre-tin, 2-3 s per joint with a 60-80 W iron). Plug W06 (UART6) and W08 (UART2 + I2C); solder W09 to the video pads and the telemetry wire of W05 to RX4.

3. ESC: solder the free end of W02 and its capacitor to the BAT pads, the motor lead W03 to the motor pads; fit the 1-pin Dupont housings of W05 (signal, GND) for S1.
4. Continuity and insulation, unpowered: + to - > 1 kOhm (capacitor charging is normal), no short between any motor phase and -, XT30 polarity matches the pack.
5. Smoke stopper first power-up (current-limited supply 25 V / 0.5 A, or a smoke stopper on the pack): FC boots, ESC beeps, no heat anywhere. Then the pack directly.
6. Fit in F2: FC on the bay floor on its 0.5 mm foam pad (arrow forward), ESC on the FC's top on foam tape, receiver on the floor ahead of the FC. Battery lead and ESC feed in the right lane (y +11), video at y +3, GPS at y -3, ESC signal on the FC's left top.
7. Motor: W03 from the ESC's rear, low on the left of the wing bay to the MR30 ahead of the tail joint; W04 through the joint's upper quadrant to the motor bulkhead.
8. Servos: aileron lead W11 under the floor to S3; tail servo leads through the joint to the JR plug pairs beside the aileron servo, extensions W12 to S4 (left) and S5 (right).
9. Strain relief: Kapton tape at every lane change, nothing resting on the motor phases or the ESC, a slack loop on W08 for the hatch, MR30 and the servo plugs reachable through the tail joint.

8. Power-up and tests

Test	Procedure / pass criterion
Voltage	Read the pack voltage with a meter and in ArduPilot; adjust BATT_VOLT_MULT until they agree within 0.05 V.
Current	Run the motor (prop off) at a steady setting with a clamp meter or a watt meter on the pack lead; adjust BATT_AMP_PERVLT.
Motor direction and rpm	Prop off: short burst, check direction (pusher) and that ESC rpm appears in ArduPilot (bidirectional DShot).
Servos	Direction and travel of each surface against the ArduPilot document; no binding at full deflection; aileron reflex 3 deg TE up at neutral.
Radio and failsafe	Bind the RP1 V2; switch the transmitter off: RTL after 5 s. Range check at reduced power.
GPS and compass	3D fix outdoors; compass calibration with the ESC and battery installed (current drawn near the compass).
Thermal	Static full throttle (prop on, aircraft held) for at most 10 s: ESC, wires, XT30 and MR30 hand-warm afterwards (connectors are near their ratings).
Video	Camera image and OSD (voltage, current, ground speed) in the goggles; VTX power and band legal for the event.

9. Open items (decide before ordering or flying)

- XT30 rating: Amass XT30 is rated 15 A continuous / 30 A peak; this aircraft draws 51 A on a pass and 60 A at static full throttle. FPV builds often run XT30 this hard for seconds, but it is outside the rating. Options: an XT60 lead with ready-made packs that have XT60 (the connector pair needs about 16 x 8 mm behind the pack instead of 10 x 5 mm), or keep XT30 and limit full power to the short passes. Decide before ordering packs.
- Series packs: with 6S 680 mAh the motor current runs through every pack's own XT30 and the series link, not only the aircraft's XT30.
- MR30: the motor phases carry about 42 A RMS on a pass, just over the MR30's 40 A one-minute rating: keep full power to the short passes; 3.5 mm gold bullets (one per phase) are the alternative.
- FC power pad positions: the routes end at the FC's front face; the cut lengths include 25-30 mm for reaching the pads wherever they sit on the board.

10. Tools and consumables

- Temperature-controlled iron 60-80 W with a chisel tip (14 AWG pads), fine tip for 28 AWG; leaded solder 0.5-0.8 mm; no-clean flux.
- Silicone wire: 14 AWG red + black, 16 AWG black (3 phases), 28 AWG in four colours.
- Heat shrink 1.5, 3, 4.8 mm; Kapton tape; double-sided foam tape; XT30 pair, MR30 pair, 1-pin Dupont housings with crimps, JST-GH 4-pin plug (supplied with the FC).
- Multimeter, smoke stopper or current-limited bench supply, clamp or watt meter for the current calibration.